

Vamshi J

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PROFESSIONAL SUMMARY

Senior **Snowflake Data Engineer** with 10+ years of experience designing enterprise-scale cloud data warehouse platforms across banking, insurance, telecom, and financial services domains. Deep expertise in **Snowflake (Streams, Tasks, Snowpipe, Secure Data Sharing, Virtual Warehouses)**, **dbt-driven ELT frameworks**, advanced **SQL optimization**, and dimensional & **Data Vault 2.0 modeling**. Proven ability to define ingestion standards, implement governance models, optimize cost-performance balance, and deliver audit-ready analytical platforms supporting regulatory and risk reporting.

TECHNICAL SKILLS

Snowflake Platform: Virtual Warehouses, Streams, Tasks, Snowpipe, External Stages (S3/ADLS), Secure Data Sharing, Time Travel, Zero-Copy Cloning, Clustering Keys, Materialized Views, Query Profile

Data Engineering: dbt, Python, Advanced SQL (Window Functions, CTEs, Query Optimization), Incremental ELT, CDC Merge Patterns

Architecture & Modeling: Star Schema, Snowflake Schema, Data Vault 2.0, Data Mart Design, Metadata-driven ELT Frameworks

Orchestration: Apache Airflow, Snowflake Tasks, Event-driven Ingestion

Cloud & Storage: AWS S3, Azure ADLS Gen2, Snowpipe Auto-Ingest

Governance & Security: RBAC Design, Row-Level Security, Masking Policies, Secure Views, Data Classification, Audit Controls

DevOps & Automation: Terraform, CI/CD (Jenkins, GitHub Actions), Git

Monitoring & Optimization: Warehouse Sizing Strategy, Resource Monitors, Cost Governance, Query Performance Tuning

PROFESSIONAL EXPERIENCE

Senior Data Engineer – Enterprise Risk Data Warehouse, Citizens Bank April 2023–Present | *Johnston, RI*

- Designed enterprise-scale **Snowflake warehouse architecture** integrating 20+ upstream banking systems into curated risk and compliance data marts, defining standardized ELT ingestion patterns using **Snowflake Streams, Tasks, and Snowpipe** to enable incremental CDC processing and eliminate full-rebuild dependency across regulatory workloads.
- Architected dimensional and **Data Vault 2.0 models** within Snowflake to support regulatory reporting and fraud analytics, implementing optimized **Clustering Keys, Materialized Views, and Micro-partition pruning** strategies that reduced complex aggregation latency by **30%** during peak reporting cycles.
- Established warehouse performance and cost governance standards by tuning **Virtual Warehouse sizing, Auto-suspend policies, and Resource Monitors**, lowering compute spend by **22%** while maintaining SLA-aligned performance for executive dashboards.
- Analyzed **Snowflake micro-partition metadata and pruning behavior** using Query Profile and clustering depth metrics, redesigning clustering keys and data distribution strategy to improve selective query pruning and reduce scan volume across regulatory workloads.
- Led migration from legacy ETL frameworks into a **dbt-driven transformation architecture** integrated with Snowflake, standardizing modular SQL models, CI/CD deployment pipelines, and version control to improve developer productivity and transformation reliability.
- Implemented enterprise security model leveraging **RBAC hierarchies, Masking Policies, Row-Level Security, Secure Data Sharing, and Time Travel**, strengthening audit compliance posture across risk and finance domains while mentoring junior engineers on Snowflake optimization practices.

Senior Data Engineer – Analytics Modernization, Pfizer April 2021–Feb 2023 | *Sanford, NC*

- Led Snowflake warehouse consolidation initiative integrating **AWS S3 external stages and Snowpipe auto-ingest**, enabling structured ingestion of multi-terabyte compliance datasets into curated Snowflake schemas aligned with finance and regulatory reporting requirements.
- Designed and implemented incremental ELT pipelines using **Snowflake Streams and MERGE logic**, improving reporting turnaround by **25%** while eliminating redundant processing across recurring analytics

workloads.

- Developed advanced transformation logic using **Snowpark (Python)** to execute in-warehouse processing for complex enrichment and validation workflows, reducing data movement between application and warehouse layers while improving transformation maintainability.
- Optimized analytical SQL queries using **Snowflake Query Profile analysis**, refining clustering strategies and warehouse concurrency scaling to significantly improve dashboard responsiveness for compliance stakeholders.
- Orchestrated end-to-end workflows using **Apache Airflow**, coordinating Snowpipe ingestion, dbt runs, validation checks, and Snowflake Tasks to increase pipeline stability and reduce operational escalations.

Data Engineer –Insurance Data Modernization, Plymouth Rock Assurance Jan 2019 – March 2021 | Boston, MA

- Designed and deployed scalable Snowflake ingestion pipelines leveraging **S3 External Stages, Snowpipe auto-ingest, and staged bulk loading**, migrating over 30M structured insurance records into curated warehouse layers with **99% validation accuracy**.
- Architected **Star Schema and Snowflake Schema data marts** supporting underwriting and claims analytics, implementing optimized **Clustering Keys and Micro-partition strategies** that improved BI reporting responsiveness by **28%**.
- Defined metadata-driven ELT standards using **Incremental MERGE patterns and dbt-based transformation layers**, reducing ETL runtime by **30%** while enabling reusable onboarding of new source systems.
- Implemented secure governance model leveraging **RBAC roles, Masking Policies, Secure Views, and Access Control frameworks**, ensuring controlled exposure of sensitive claims and financial datasets across business units.
- Designed ingestion pipelines handling semi-structured JSON datasets using **Snowflake VARIANT columns, FLATTEN functions, and schema evolution strategies**, enabling scalable processing of nested AML and transaction event payloads.
- Automated Snowflake object migrations and environment promotions using **Terraform and CI/CD pipelines**, standardizing deployment practices and reducing release-related production defects.

Data Engineer-Telecom Data Platform, Verizon Feb 2017 – Dec 2018 | Irving, TX

- Developed large-scale SQL-based transformation workflows preparing telecom event datasets for centralized warehouse ingestion, supporting billing and churn analytics across high-volume subscriber environments.
- Optimized complex aggregations and partition logic within distributed processing systems, reducing heavy reporting runtime by **30%** during peak billing cycles.
- Contributed to early cloud data warehouse adoption initiatives, preparing structured datasets for integration into scalable cloud-based reporting platforms.
- Strengthened reconciliation and validation frameworks to improve reporting accuracy across distributed datasets.
- Collaborated cross-functionally with analytics and operations teams to align reporting logic with business KPIs and operational definitions.

Data Engineer - Financial Reporting Systems -HSBC GLT June 2013 – Nov 2016 | Hyd, India

- Developed and maintained SQL-driven ETL pipelines supporting regulatory, risk, and finance reporting across high-volume trading and transaction processing environments.
- Optimized complex aggregation and reconciliation logic used in month-end cycles, improving reporting performance by **25%** and reducing deadline-driven delays.
- Designed validation and balancing controls across trading/settlement datasets, reducing regulatory reporting discrepancies by **30%** during audit cycles.
- Refactored legacy batch scripts into modular SQL/Hive workflows, improving maintainability and reducing operational risk during change windows.
- Supported performance and stability of on-prem Hadoop workloads (Hive/HBase) to ensure consistent query behavior under peak reporting demand.
- Automated job sequencing and recovery patterns, reducing operational escalations by **20%** during critical

reporting windows.

- Worked directly with finance/risk/compliance stakeholders to interpret reporting requirements and deliver governed, audit-ready datasets with consistent definitions.

EDUCATION

Bachelor of Technology, JNTUH
Computer Science

2009 – 2013 | Hyderabad, India